SIM2001 Advanced Calculus : Tutorial 4

1. Use Taylor's formula for $f(x, y)$ at the origin to find quadratic and cubic approximations of f near the origin.

   (13.9.3) $f(x, y) = y \sin x$

   (13.9.5) $f(x, y) = e^x \ln(1 + y)$

2. Evaluate the iterated integral.

   (14.1.5) $\displaystyle\int_0^3 \int_0^2 (4 - y^2) dy dx$

   (14.1.7) $\displaystyle\int_0^1 \int_0^1 \frac{y}{1 + xy} dx dy$

   (14.1.11) $\displaystyle\int_{-1}^2 \int_0^{\pi/2} y \sin x dx dy$

3. Evaluate the double integral over the given region R.

   (14.1.19) $\displaystyle\iint_R xy \cos y dA, \qquad R : -1 \le x \le 1,\ 0 \le y \le \pi$

   (14.1.23) $\displaystyle\iint_R \frac{xy^3}{x^2 + 1} dA, \qquad R : 0 \le x \le 1,\ 0 \le y \le 2$

4. (14.1.29)
   Find the volume of the region bounded above by the paraboloid $z = x^2 + y^2$ and below by the square $R : -1 \le x \le 1,\ -1 \le y \le 1$.

5. Sketch the region of integration and evaluate the integral.

   (14.2.23) $\displaystyle\int_0^1 \int_0^{y^2} 3y^3 e^{xy} dx dy$ [(14.2.31) in 15th Ed.]

6. Integrate f over the given region.

   (14.2.25) [(14.2.33) in 15th Ed.]
   $f(x, y) = \dfrac{x}{y}$ over the region in the first quadrant bounded by the lines $y = x$, $y = 2x$, $x = 1$, $x = 2$.

7. Sketch the region of integration, reverse the order of integration and evaluate the integral.

   (14.2.47) $\displaystyle\int_0^\pi \int_x^\pi \frac{\sin y}{y} dy dx$ [(14.2.55) in 15th Ed.]

   (14.2.53) $\displaystyle\int_0^{1/16} \int_{y^{1/4}}^{1/2} \cos(16\pi x^5) dx dy$ [(14.2.61) in 15th Ed.]

   (14.2.55) [(14.2.63) in 15th Ed.]
   $\displaystyle\iint_R (y - 2x^2) dA$ where R is the region bounded by the square $|x| + |y| = 1$

8. (14.2.57) [(14.2.65) in 15th Ed.]
   Find the volume of the region bounded above by the paraboloid $z = x^2 + y^2$ and below by the triangle enclosed by the lines $y = x$, $x = 0$ and $x + y = 2$ in the xy-plane.

9. (14.2.59) [(14.2.67) in 15th Ed.]
   Find the volume of the solid whose base is the region in the xy-plane that is bounded by the parabola $y = 4 - x^2$ and the line $y = 3x$, while the top of the solid is bounded by the plane $z = x + 4$.

10. (14.2.61) [(14.2.69) in 15th Ed.]
Find the volume of the solid in the first octant bounded by the coordinate planes, the plane $x = 3$, and the parabolic cylinder $z = 4 - y^2$.

11. Sketch the region bounded by the given lines and curves. Then express the region's area as an iterated double integral and evaluate the integral.

(14.3.7) The parabolas $x = y^2$ and $x = 2y - y^2$

(14.3.11) The lines $y = 2x$, $y = \dfrac{x}{2}$ and $y = 3 - x$